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Master Thesis

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Author:
Lona Frießner

Supervisor:
Prof. Dr. Simon Grund

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Abstract

Institute of Psychology

Department of Psychology with focus on Quantitative Methods

Master of Science

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by Lona Frießner

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2.1 Theoretical foundation of multilevel models

Example

Imagine you are interested in the musical self-concept of young adults and whether it is related to how many hours they practice their instrument each week.
To investigate this question, you collect data from several youth ensembles. You recruit 15 ensembles, each consisting of approximately 10 players. Every player completes a questionnaire assessing their musical self-concept as well as their weekly practice time.
At first glance, this seems like a simple dataset of 150 individuals. But is it really that simple?
Players from the same ensemble rehearse together, share teachers, and experience similar social and musical environments. Their responses may therefore be more similar within ensembles than between ensembles.

Populations like this, where units of interest (e.g., music players, students, employees) are grouped into higher-level units (e.g., ensembles, classes, firms), are common in the human social life and therefore in psychological studies.
The resulting data is called *nested* or *clustered* (Snijders & Bosker, 2012, p. 7).
From a sampling perspective, a so-called *multistage sampling process* (sample groups first, then individuals inside these groups[1]) is not independent: After selecting a group, you will try to sample all of the members of this group, making it more likely to select one person in this group over others in non-selected groups (Snijders & Bosker, 2012, p. 7). More importantly, individuals inside the same group typically share contextual influences (e.g., the same ensemble teacher, the same class room) and thus will give more similar responses in surveys and behavior. As a consequence, the basic assumption of independence in traditional linear modeling is violated, because error terms will be correlated (Raudenbush & Bryk, 2010, p. xx).
One well-suited and popular possibility to account for this fact is the use of *multilevel models* (also called *hierarchical linear models* or *mixed models*).

[1]In this thesis, the term 'groups' refers to the higher-level units of the data and level-1 units are referred to as 'persons'. However, it is important to note that these models can also be used for longitudinal research, where measurement points are on level 1 and individuals are the level-2 units. Furthermore, more than two levels are also possible, but are not discussed here.

2.1.1 The concept of multilevel regression models

The aim of regression analysis, whether single- or multilevel, is to explain variance in an outcome variable by including associated variables in the model as predictors (Snijders & Bosker, 2012, p. 80). In single-level regression, the residual e_i captures the *unexplained variance* in Y, that is differences in the outcome from the prediction that the model made (Leonhart, 2013, p. 319). The assumption of the classical linear model is that after inclusion of all *systematic* influences on Y, the residuals are *random*. Specifically, they are assumed to have expectation zero ($E(e_i) = 0$), constant variance across observations (homoscedasticity; $Var(e_i) = \sigma^2$), and to be independent across observations ($Cov(e_i, e_{i'}) = 0$ for $i \neq i'$) (Fahrmeir et al., 2021, p. 87).

Multilevel models (MLM) extend regular linear regression by adding level-2 *random effects* to explain the variability between groups that cannot be accounted for by predictors (Snijders & Bosker, 2012, p. 80). This partition of variance can be expressed in the multilevel regression model without predictors, the *empty model*:

$$Y_{ij} = \gamma_{00} + u_{0j} + e_{ij}$$

Here, the outcome value of person i in group j is modeled as the sum of the overall mean γ_{00}, a group level random effect u_{0j} and a random effect on the individual level, e_{ij} (Snijders & Bosker, 2012, p. 49). The random intercept u_{0j} captures the dependency of individuals within one group statistically, as all individuals in group j have the same u_{0j} value and their outcome values are therefore correlated. Conditional on this group component, however, the remaining level-1 residuals e_{ij} are assumed to be independent, addressing the issue of dependency that would arise if a single-level regression model were applied to nested data.

This and the further explained models are illustrated in Figure 2.1.

2.1.2 Within- and between- group regressions and inclusion of predictors

Predictors can then be included to explain variability within groups (level 1) and between groups (level 2).
At level 1, the outcome of a person i in group j is modeled as a function of individual-level explanatory variables X_{ij} to obtain the within-group regression model. For simplicity, the model is illustrated with a single level-1 predictor:

$$Y_{ij} = \beta_{0j} + \beta_{1j}X_{ij} + e_{ij} \tag{2.1}$$

At level 2, the variation between groups in the group-dependent coefficients β_{0j} and β_{1j} is modeled. Each group may have its own intercept and slope. These coefficients can be separated into a fixed part (the average coefficient across groups) and a random deviation from this average (Snijders & Bosker, 2012, p. 75). Without level-2 predictors:

$$\beta_{0j} = \underbrace{\gamma_{00}}_{\text{Fixed part}} + \underbrace{u_{0j}}_{\text{Random part}}$$

$$\beta_{1j} = \underbrace{\gamma_{10}}_{\text{Fixed part}} + \underbrace{u_{1j}}_{\text{Random part}}$$

On this level, in order to explain between-group variability in the intercepts and slopes, level-2 predictors Z_j may be included (Snijders & Bosker, 2012, p. 80). Including such predictors may reduce the unexplained variance of the random effects u_{0j} and u_{1j}. With a single level-2 predictor, the model becomes:

$$\beta_{0j} = \gamma_{00} + \gamma_{01} Z_j + u_{0j} \tag{2.2}$$

$$\beta_{1j} = \gamma_{10} + \gamma_{11} Z_j + u_{1j} \tag{2.3}$$

Substituting the level-2 equations Equation 2.2 and Equation 2.3 into the level-1 model Equation 2.1 yields the combined multilevel model:

$$Y_{ij} = \underbrace{\gamma_{00} + \gamma_{10} X_{ij} + \gamma_{01} Z_j + \gamma_{11} X_{ij} Z_j}_{\text{Fixed part}} + \underbrace{u_{0j} + u_{1j} X_{ij} + e_{ij}}_{\text{Random part}}$$

Y_{ij}: Outcome for individual i in group j
X_{ij}: Level-1 (within-group) predictor
Z_j: Level-2 (between-group) predictor

γ_{00}: Fixed intercept (overall mean when $X_{ij} = 0$, $Z_j = 0$)
γ_{10}: Average within-group effect of X_{ij} on Y_{ij}
γ_{01}: Effect of Z_j on the intercept (between-group effect)
γ_{11}: Cross-level interaction; change in the effect of X_{ij} depending on Z_j

u_{0j}: Random intercept (group-specific deviation from γ_{00})
u_{1j}: Random slope (group-specific deviation from γ_{10})
e_{ij}: Level-1 residual

If no level-2 predictors are included ($Z_j = 0$), the model reduces to the simpler random-coefficients model without contextual predictors.

2.1.3 Distributional assumptions of the multilevel regression model

Similar distributional assumptions are made for the level-2 random effects as for the residuals in single-level regression: they have mean zero, constant variances across groups, and are independent across clusters and independent of the level-1 residuals. However, random effects within the same cluster may be correlated and are often expected to do so (Snijders & Bosker, 2012, p. 76). Additionally, residuals and random effects are commonly assumed to follow a normal distribution for convenience, as it facilitates the likelihood-based estimation of the unknown parameters (Fahrmeir et al., 2021, p. 388). In summary:

$$\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \mathbf{T} \right), \qquad \mathbf{T} = \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{01} & \tau_1^2 \end{pmatrix}$$

$$e_{ij} \sim \mathcal{N}(0, \sigma^2)$$

$$\mathrm{Cov}(u_{0j}, e_{ij}) = 0, \quad \mathrm{Cov}(u_{1j}, e_{ij}) = 0$$

2.1.4 The random intercept model

In many empirical applications of multilevel models the slope β_{1j} is assumed to be constant across groups ($u_{1j} = 0$), leading to the *random intercept model* (Snijders & Bosker, 2012, p. 45). This model allows for baseline differences in the outcome between groups, whereas the strength of the relationship between predictors and the outcome is assumed to be constant:

$$Y_{ij} = \gamma_{00} + \gamma_{10} X_{ij} + \gamma_{01} Z_j + u_{0j} + e_{ij}$$

The random intercept model also provides a convenient framework for examining the use of *group means* of a given level-1 predictor as additional level-2 predictors. Including the group mean as a contextual variable allows the within-group regression coefficient to vary from the between-group regression coefficient (Snijders & Bosker, 2012, p. 57). This makes the model more flexible and often more realistic, as so-called *contextual effects* are frequently observed in empirical settings (Snijders & Bosker, 2012, p. 27). In such cases, the effect of an individual-level predictor differs from the effect of the average value of this predictor. Example ... illustrates how the processes operating within groups may differ from those operating between groups. To separate within- and between-group effects, the level-1 predictor is then often *group-mean-centered*: Instead of using X_{ij} directly, the centered predictor $(X_{ij} - \bar{X}_j)$ is entered into the model. This decomposes within- and between- group effects and facilitates their interpretation (Snijders & Bosker, 2012, p. 58). A random intercept model including such a decomposition can be written like this:

$$Y_{ij} = \gamma_{00} + \gamma_{10}(X_{ij} - \bar{X}_{.j}) + \gamma_{01}\bar{X}_{.j} + u_{0j} + e_{ij} \tag{2.4}$$

In this random intercept model with context effect, musical self-concept of student i in ensemble j is predicted by the sum of γ_{00}: the average musical self-concept across all ensembles $X_{ij} - \bar{X}_{.j}$: the group-mean centered weekly practice duration of this student $\bar{X}_{.j}$: the average practice duration of this student's ensemble u_{0j}: a random deviation of ensemble j's average MSC from the overall mean (unexplained differences between ensembles) e_{ij}: a random deviation of the student's MSC from the predicted value within their ensemble
The two regression coefficients represent the following effects: γ_{10}: within-group effect, indicating the influence of a student's own WPD on their MSC, controlling for the ensemble context γ_{01}: the contextual effect, indicating the influence of the ensemble's average WPD on the student's MSC

For the remainder of this thesis, the focus lies on this particular random intercept model.

2.1.5 The intraclass correlation coefficient (ICC)

In the random intercept model, the variance components are given by

$$u_{0j} \sim \mathcal{N}(0, \tau_0^2), \qquad e_{ij} \sim \mathcal{N}(0, \sigma^2), \qquad \mathrm{Cov}(e_{ij}, u_{0j}) = 0.$$

From these variance components, the proportion of variability that is attributable to between-group differences can be calculated. This quantity is referred to as the *intraclass correlation coefficient* (ICC). In the empty model, the ICC is equal to the correlation between the outcomes of two randomly chosen individuals from the same group (Snijders & Bosker, 2012, p. 18). In models including predictors, the ICC is

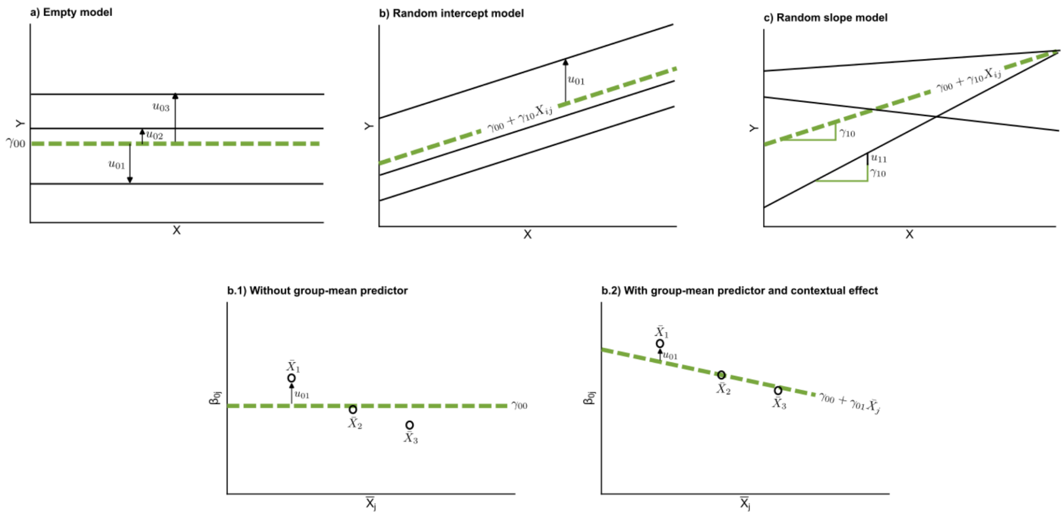


Illustration of multilevel model specifications. Empty model, random intercept model and random slope model on Level 1. For level 2, the difference between a random intercept model with and without group-mean predictor is shown.
Erklärung der Plots

FIGURE 2.1

interpreted conditionally on the predictors, that is, as the correlation between two individuals from the same group who share identical predictor values (also referred to as the *residual ICC*) (Snijders & Bosker, 2012, p. 52).

Formally, the ICC is defined as

$$ICC = \frac{\tau_0^2}{\tau_0^2 + \sigma^2}$$

The ICC is an important diagnostic metric because it indicates whether the use of a multilevel model is advantageous. If $ICC = 0$, no between-group variance is present and individuals within the same group are not more similar to each other than to individuals from different groups (conditional on the predictors included in the model). In this case, the independence assumption of classical regression is not violated and a conventional single-level regression model is sufficient.
Conversely, larger ICC values indicate increasing similarity within groups and thus statistical dependency of observations. Ignoring this dependency and applying a classical linear model leads to biased standard errors, confidence intervals, and hypothesis tests (Fahrmeir et al., 2021, p. 370).

2.2 Estimation of multilevel models

In the preceding chapter, the theoretical foundations and parameters of the multilevel model were introduced. In practice, however, these parameters are unknown and only exist in the theoretical realm as 'true' values. The fixed effects γ and variance components σ^2 and τ^2 must therefore be estimated from observed data. Based on these estimates, inferences about the underlying population parameters are drawn.
This process is illustrated for the random intercept model in Figure 2.2. Provided that the model specification corresponds to the true data-generating process well

enough, the reliable performance of this 'inferential machinery' depends on three main elements (Elff et al., 2021):

1. Estimation of fixed-effect coefficients (regression parameters)
2. Estimation of variance–covariance components of the random effects (u) and residual errors (e)
3. Selection of an appropriate approximation of the sampling distribution of the test statistic.

Aspects of the observed data may pose challenges to this estimation process. Moreover, the inferential machinery itself (i.e., its formulas and underlying assumptions) may be inapt for the empirical data reality. Two common challenging aspects, namely small sample sizes and missing data, will be discussed in the following chapters.

2.3 Small sample sizes in multilevel models

2.3.1 Frequentist corrections for small samples

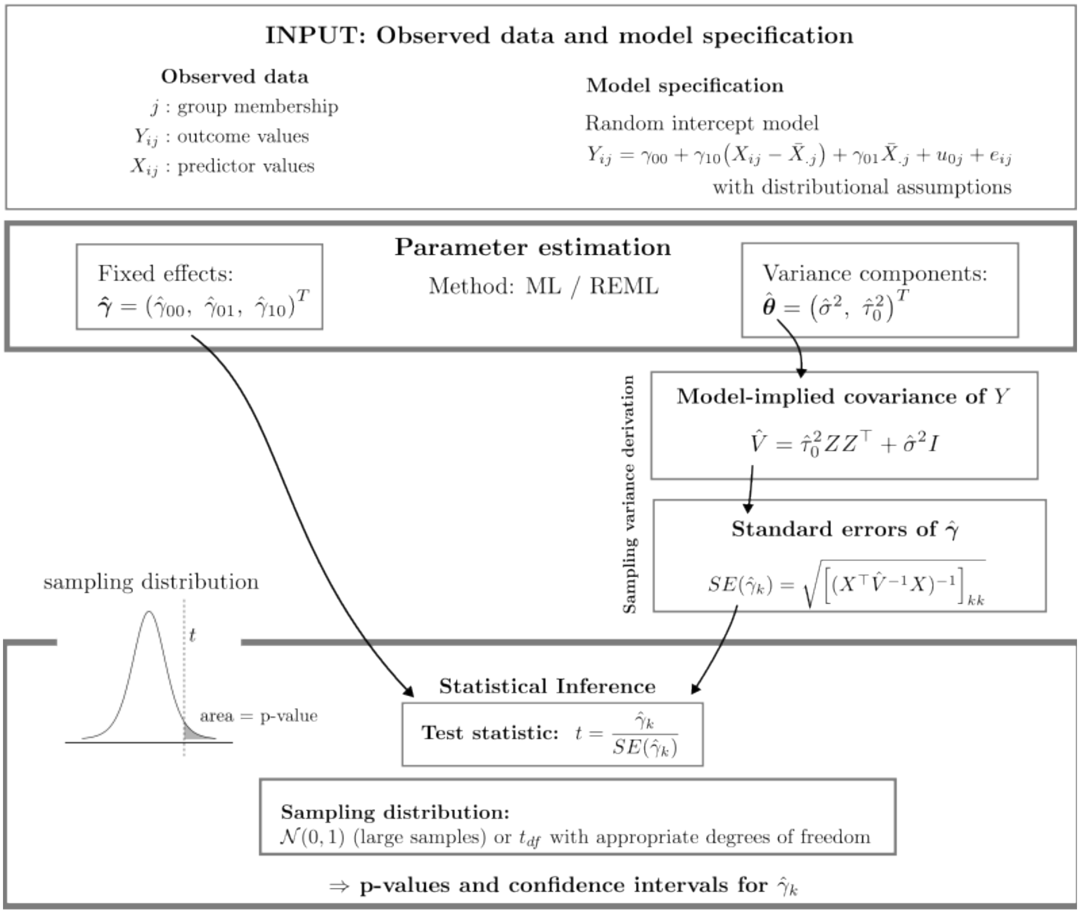
Frequentist estimation procedures of linear multilevel models are typically based on asymptotic theory, meaning that biased estimates may occur in small samples. In multilevel models, the number of groups is particularly important. Depending on the model of interest, roughly 25-30 clusters are found to be necessary to obtain precise results (D. McNeish, 2017a; D. M. McNeish & Stapleton, 2016). In practice, however, these numbers are often difficult to achieve, as recruiting a large number of groups can be difficult and costly. The main issues associated with small-sample multilevel models and corresponding solutions are presented below.

2.3.1.1 Estimation of regression parameters

The regression parameters are estimates of the effects of interest. For example, how much does one additional hour of weekly practice influence a student's musical self-concept? How does the musical self-concept differ between ensembles that practice on average three hours per week and those that practice thirteen hours per week?
Kackar & Harville (1981) showed that both ML and REML estimators of variance components are even and translation-invariant, implying that fixed effect point estimates are unbiased, independent of sample size.

2.3.1.2 Estimation of variance components

The estimation of the variance components is more delicate. In ML estimation, fixed effects and variance components are estimated simultaneously, typically via iterative procedures like the Expectation Maximization (EM) mechanism (D. McNeish, 2017b). In this process fixed effects are effectively being treated as known when estimating variance components. Consequently, the sampling variability of the fixed-effect estimates is ignored and the loss of degrees of freedom due to their estimation is not accounted for. While these issues are not troublesome in large samples, they lead to systematic underestimation of variance components in small samples (D. McNeish, 2017b).
REML fixes this issue by separating the estimation conceptually. The fixed effects are eliminated from the likelihood, and variance components are estimated solely based on residual information. Fixed effects are then estimated via generalized least squares (GLS), using the previously estimated variance components (Raudenbush & Bryk,



Illustrative example. For a hypothetical data set with four persons nested in two groups:

The corresponding model matrices are $X = \begin{pmatrix} 1 & 2 \\ 1 & 1 \\ 1 & 3 \\ 1 & 5 \end{pmatrix}$ $Z = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}$ $Y = \begin{pmatrix} 5 \\ 4 \\ 2 \\ 5 \end{pmatrix}$

and the implied covariance matrix is
$$\hat{V} = \begin{pmatrix} \hat{\tau}_0^2 + \hat{\sigma}^2 & \hat{\tau}_0^2 & 0 & 0 \\ \hat{\tau}_0^2 & \hat{\tau}_0^2 + \hat{\sigma}^2 & 0 & 0 \\ 0 & 0 & \hat{\tau}_0^2 + \hat{\sigma}^2 & \hat{\tau}_0^2 \\ 0 & 0 & \hat{\tau}_0^2 & \hat{\tau}_0^2 + \hat{\sigma}^2 \end{pmatrix}$$

Illustrative observed values

id	j	X_{ij}	Y_{ij}
1	A	2	5
2	A	1	4
3	B	3	2
4	B	5	5

Simplified frequentist inferential process in a random intercept model, based on Snijders & Bosker (2012) and Raudenbush & Bryk (2010).

Notes on notation

- X: design matrix for the fixed effects. Each row corresponds to an observation (e.g., a person) and each column to a predictor included in the fixed part of the model.
- Z: design matrix for the random effects. In the random intercept model it encodes group membership (1 = observation belongs to the group, 0 = otherwise), so observations from the same group share the same random intercept.
- Y: outcome vector containing the observed values of the dependent variable.
- $\hat{V}$: estimated variance–covariance matrix of the outcome vector Y. In the random intercept model it is determined by the estimated variance components of the random intercept ($\hat{\tau}_0^2$) and the residual variance ($\hat{\sigma}^2$), reflecting that observations within the same group are correlated.
- $\hat{\gamma}_k$: the k-th element of the estimated fixed-effect vector $\hat{\gamma}$.

FIGURE 2.2

2010, p. 410). This leads to less downward-biased variance component estimates, while ML and REML become asymptotically equivalent as sample size increases.

Nevertheless, the REML variance estimate of the fixed effects may still be too small in small samples for two reasons. First, the variance formula treats the estimated variance components $\hat{\theta}$ as known and therefore ignores their sampling variability (Kackar & Harville, 1984). Second, the REML variance expression itself relies on asymptotic approximations (D. McNeish, 2017b). Both issues can lead to underestimated standard errors. Several small-sample corrections have been proposed to address this problem, among which the Kenward–Roger procedure is one of the most widely used (D. McNeish, 2017b).

2.3.1.3 Approximation of the sampling distribution

The standardized test statistic used for inference (see Figure 2.2) can be regarded as 'small-sample-adjusted' at this point. The final step concerns the choice of a sampling distribution for p-values and confidence intervals. In small samples, a t-distribution is typically recommended instead of a standard normal distribution (Snijders & Bosker, 2012, p. 95).

The shape of the t-distribution is determined by the number of degrees of freedom (HIER BILDER VON VERSCHIEDENEN T VERTEILUNGEN?). With small degrees of freedom, even small changes noticeably affect the shape of the sampling distribution and therefore the resulting p-values and confidence intervals. The concept of degrees of freedom is difficult to define concisely, but it can be described as the number of independent pieces of information in the data that remain after accounting for restrictions such as estimated parameters (Pandey & Bright, 2008).

In multilevel models, the degrees of freedom generally cannot be calculated exactly and must therefore be approximated because of the nested data structure (Snijders & Bosker, 2012, p. 95). The Kenward–Roger correction mentioned earlier also includes a complex small-sample-adjusted approximation of the degrees of freedom, building on the Satterthwaite method (D. McNeish, 2017b).

A more straightforward approximation is provided in textbooks by Raudenbush & Bryk (Raudenbush & Bryk, 2010, p. 58) and Snijders & Bosker (Snijders & Bosker, 2012, p. 95), and has also been proposed in methodological articles (Elff et al., 2021). This approach distinguishes between level-1 and level-2 predictors and reflects the concept of degrees of freedom as information minus estimated parameters:

$$df_{Lv1} = N_1 - r - 1, \qquad df_{Lv2} = N_2 - q - 1$$

N_1 : Level-1 sample size (total number of individuals)
r : Total number of predictors
N_2 : Level-2 sample size (number of groups)
q : Number of predictors at level two

2.3.2 Fully Bayesian analysis

As alternative to the frequentist framework predominantly presented here, a Bayesian approach is often recommended for small sample sizes (D. McNeish, 2016). In this framework, parameters are viewed as random variables with probability distributions. Prior knowledge about the parameters of interest in the form of prior distributions is combined with the information in the observed data in the form of a likelihood function. Together they are combined via Bayes' theorem to form the posterior distribution of the parameter of interest, namely the updated knowledge and certainty about the parameters (Van De Schoot et al., 2014). More general details on Bayesian analyses will not be discussed in this thesis.

Two aspects of Bayesian analyses are particularly helpful in small samples.
A Bayesian framework is naturally equipped to account for the additional uncertainty arising in small samples because it does not rely on large-sample asymptotic approximations (D. McNeish, 2016). Instead, the posterior distribution, the end result of a Bayesian analysis, is typically approximated using Markov Chain Monte Carlo (MCMC) sampling. Consequently, Bayesian inference does not require small-sample corrections for standard errors or degrees of freedom.
Furthermore, Bayesian inference combines prior information about parameters with the observed data. This makes it possible to include previous knowledge into the analysis to increase the amount of available information (D. McNeish, 2016). This advantage, however, depends strongly on prior specification. In small samples, prior distributions can have a substantial influence on the results because the likelihood function is often relatively flat and therefore contains limited information about the parameters (Gelman et al., 2025, p. 355). Informative or weakly informative priors can improve estimation accuracy, whereas vague or software default priors may be more influential than intended and can perform poorly in small samples (Gelman, 2006; D. McNeish, 2016; D. M. McNeish, 2016). Nevertheless, even Bayesian estimation with noninformative priors has in some settings been found to outperform maximum likelihood estimation in terms of RMSE (Zitzmann et al., 2015). In addition, prior distributions may prevent the estimation of inadmissible values that could otherwise lead to convergence problems, such as negative variance estimates (Van De Schoot et al., 2014). Thus, the inclusion of prior distributions may be regarded either as a methodological advantage or as "a price to be paid" (Raudenbush & Bryk, 2010, p. 411) for Bayesian inference.

2.4 Missing data in multilevel models

> Example
>
> In the study about musical self-concept, you find after data collection that some students in several ensembles did not report how many hours they practice per week. They completed the questionnaire on musical self-concept but left the practice question blank.
> The reason for this is unknown: they might have forgotten to answer the question, skipped it intentionally, or misunderstood it. Regardless of the reason, the data set now contains missing values. How should these missing observations be handled in the analysis?

Missing data (i.e., incomplete data sets) are a widespread issue in empirical research and of course also applies to multilevel data, e.g., because individuals drop out of studies or forget to fill out certain items of a questionnaire. Three major concerns commonly arise from data that contain missing values (Lüdtke et al., 2007):

1. The effective sample size is reduced, leading to lower efficiency in estimation (this is especially troublesome if the would-be sample size is already small)

2. Typical statistical methods are difficult to conduct, because they were designed for complete data.
3. Parameter estimation may be biased, if the missing and the observed data differ from each other systematically

Therefore, it is crucial to treat missing data appropriately. In the last three decades, this issue has received attention in psychological research, and modern methods were developed (Enders, 2022; Schafer & Graham, 2002), more recently also for the more complex case of multilevel data (Enders, 2025; Lüdtke et al., 2017). This chapter discusses these methods, after giving a short introduction into different missing data mechanisms and patterns.

2.4.1 Missing data mechanisms

Why data are missing can have multiple reasons, and different mechanisms may lead to different consequences in statistical inference. Rubin and Little (Little & Rubin, 1987; Rubin, 1976) introduced a typology and theoretical framework of missing data mechanisms that describes the relationship between the observed and the missing data. The missing data mechanism is generally not empirically testable from the observed data alone and assumptions about the mechanism must therefore be theoretically justified (Enders, 2022, p. 14ff.).

Missing completely at random (MCAR). Considering the hypothetical complete data matrix Y[2], Y_{com} is partitioned into observed, Y_{obs}, and missing parts, Y_{mis}, and R denotes the missingness[3]. Missing values are considered to be MCAR if the distribution of missingness does not depend on the data (Schafer & Graham, 2002):

$$P(R \mid Y_{com}) = P(R)$$

In other words, no variable (observed or unobserved) can predict missingness and every individual in every group has the same chance of missing values (Enders, 2022, p. 6).

Missing at random (MAR). The MAR mechanism describes the case when the probability of missingness does depend on observed data, but not on missing values (Enders, 2022, p. 8):

$$P(R \mid Y_{com}) = P(R \mid Y_{obs})$$

Missingness is therefore not random, as the name would suggest, but *conditionally random*: After controlling for the observed data, the missingness is purely random (Enders, 2022, p. 8).

For example, students who practice less might be more likely to skip the question about weekly practice duration. If this behavior can be predicted from other observed variables (musical self-concept or ensemble membership), the MAR assumption could be plausible.

[2] Here, Y represents the complete data set, including both outcome variables and predictors.

[3] R is treated as a set of random variables that indicate for each value if it is missing or not (Schafer & Graham, 2002).

Missing not at random (MNAR). If missingness depends on unobserved values, this is called a missing not at random process[4]:

$$P(R \mid Y_{com}) = P(R \mid Y_{obs}, Y_{mis})$$

Two individuals in the same group with identical scores in all observed variables therefore no longer have the same chance of missing values, because this chance depends on the unavailable, missing information itself (Enders, 2022, p. 11).

While these mechanisms also apply to multilevel data, the situation becomes more complex because missing values can occur in the outcome, in level-1 predictors, in level-2 predictors, or even in the grouping variable (or combinations of these; Buuren (2021)).

2.4.2 Methods for handling missing data

Many estimation procedures require complete data. Therefore, missing data must be handled explicitly before model estimation. If researchers do not actively specify a method, statistical software often defaults to using only complete observations (Lüdtke et al., 2007). This simple ad hoc procedure is called *listwise deletion* (LD) or *complete case analysis* and is a widely used approach to handling missing data. Cases with at least one missing value are discarded, and the model is estimated using only complete observations (Graham, 2009). In multilevel settings, missingness on the group level implies that all individuals in this group are excluded as well.
Listwise deletion has two major disadvantages (Enders, 2022, p. 24):

1. Because partial data is unused, statistical power is reduced.

2. If the missing data mechanism is not MCAR, parameter estimates may be highly biased.

LD can still be adequate under some conditions (Enders, 2022, p. 25; Graham, 2009), but in already small samples, additional data deletion should be avoided as power is already limited (D. McNeish, 2017a).

More modern and better-performing methods are available and will be introduced in the following sections. These approaches generally rely on the assumption that data are missing at random (MAR). Methods for dealing with MNAR processes are not discussed in this thesis.

Modern approaches to missing data can be broadly specified into two classes:

1. *Model-based approaches* to missing data specify a statistical model for the complete data and base inference on the likelihood or posterior distribution implied by that model. Missing values are handled directly within the estimation procedure (Little & Rubin, 2020, p. 25).
2. *Imputation-based approaches* are two-fold: First, substitutions for the missing values are estimated, leading to complete data sets, which are then analyzed using standard methods.

[4]The special case $P(R \mid Y_{com}) = P(R \mid Y_{mis})$ is referred to as **focused MNAR** (Enders, 2022, p. 11).

2.4.2.1 Multiple Imputation (MI)

Multiple imputation is an imputation-based method, as the name suggests. The idea behind it is to first solve the missing data problem by obtaining reasonable guesses for the missing data, called imputations, and subsequently fitting the complete data analysis on the imputed data. In order to include the uncertainty due to missing values, in MI several imputations for each missing observation are generated. By incorporating the variance between these imputations, the additional uncertainty can be captured (Enders, 2022, p. 283).

2.4.2.1.1 Creating imputations

Imputations can be understood as "predicted values plus noise" (Enders, 2022, p. 266). Multiple imputation is closely related to Bayesian data augmentation techniques, in which missing values are treated as additional unknown quantities (Tanner & Wong, n.d.). From the joint distribution of regression parameters θ and the missing values $P(\theta, Y_{mis} \mid Y_{obs})$, parameters and imputations are simulated iteratively by an MCMC algorithm (Enders, 2022, p. 190). While during this process many thousand posterior draws are generated, only a small number (usually between $m = 20$ and $m = 100$ is recommended, Graham et al. (2007)) of imputations is saved, leading to m complete data sets. The distribution of missing values (i.e., the imputation model) is either conceptualized jointly for all variables (*joint imputation*) or individually (*fully conditional specification*). Both approaches are also suitable for multilevel data structures (Enders et al., 2016). In joint modeling, a single multivariate distribution for all variables with missing values is specified, often the multivariate normal distribution in case of continuous variables (Enders, 2022, p. 263). *Fully Conditional Specification* approximates the joint distribution by cycling through a sequence of univariate models for each variable with missings, conditional on the others. This approach is suitable if many ordinal or nominal variables are included in the data, as for every variable the researcher specifies a separate imputation model (Lüdtke et al., 2007).

Compatibility of imputation and analysis model A crucial point in the choice of a valid imputation model is compatibility with the focal analysis model that is used later on. That is, the imputation model must reflect the key structural features of this analysis model (Grund et al., 2018). If it fails to do so, parameter estimates may become biased because the imputed values no longer preserve the structure of the data. Consider, for example, the random intercept model with group means (Equation 2.4). In this setting, the imputation model must account for the nested data structure (i.e., random effects at the group level) and allow for different effects at the individual and the group level (Grund et al., 2018). **Auxiliary variables** Importantly, the imputation model is allowed to be 'richer' than the analysis model. Including additional effects or variables that are not part of the analysis model as *auxiliary variables*[5] is considered beneficial for parameter estimation (Collins et al., 2001). These variables could either be a cause of missingness, thereby making the assumption of missing at random (MAR) more plausible. Furthermore, they may correlate with variables that contain missing values, which increases the precision of the imputations and reduces bias in parameter estimates (Collins et al., 2001).

[5] Auxiliary variables are referred to as variables whose only purpose it is to improve the missing data imputation (Collins et al., 2001).

Tip

In the musical self-concept study, the years of teaching experience of the ensemble teachers might also be available. Although this variable is not part of the analysis model, it may correlate with practice behavior or musical self-concept and can therefore serve as an auxiliary level-2 variable.

In this thesis only the joint imputation will be considered. The *multivariate empty model* represents a suitable joint imputation model for the random intercept analysis model (Grund et al., 2016). Revisiting the example from the previous chapter, an imputation model with one auxiliary variable can be written as

$$\begin{pmatrix} Y_{ij} \\ X_{ij} \\ W_j \end{pmatrix} = \begin{pmatrix} \beta_{0Y} \\ \beta_{0X} \\ \beta_{0W} \end{pmatrix} + \begin{pmatrix} u_{Yj} \\ u_{Xj} \\ u_{Wj} \end{pmatrix} + \begin{pmatrix} e_{Yij} \\ e_{Xij} \\ 0 \end{pmatrix}$$

Here, W_j represents a level-2 auxiliary variable and therefore has no level-1 residual term. The model preserves the clustered data structure by allowing random intercepts at the group level.

The random effects and residuals are assumed to follow multivariate normal distributions

$$\begin{pmatrix} u_{Yj} \\ u_{Xj} \\ u_{Wj} \end{pmatrix} \sim N(0, \Psi), \qquad \begin{pmatrix} e_{Yij} \\ e_{Xij} \end{pmatrix} \sim N(0, \Sigma)$$

where the covariance matrix Ψ captures the between-group variability and covariation of the random intercepts and Σ represents the within-group covariance structure of the level-1 residuals:

$$\Psi = \begin{bmatrix} \psi_y^2 & \psi_{yx} & \psi_{yw} \\ \psi_{yx} & \psi_x^2 & \psi_{xw} \\ \psi_{yw} & \psi_{xw} & \psi_w^2 \end{bmatrix}, \qquad \Sigma = \begin{bmatrix} \sigma_y^2 & \sigma_{yx} \\ \sigma_{yx} & \sigma_x^2 \end{bmatrix}$$

2.4.2.1.2 Analyzing the completed data sets

After the imputation step, researchers are in the comfortable position to have m completed data sets that can be analyzed using standard frequentist methods. To each of the data sets, the substantive analysis model of interest (e.g., the random intercept model) is now fitted. Each analysis proceeds according to the standard estimation procedure described in Section 2.2, up to the computation of parameter estimates and their standard errors. Hypothesis testing is not conducted at this stage. The result of this step is a set of m parameter estimates and corresponding standard errors.

2.4.2.1.3 Pooling the results

These m estimates and standard errors are aggregated into a single set of results following Rubin's pooling rules (Rubin, 1987).
He defined the point estimates of effects as the arithmetic average of the m estimates:

$$\hat{\theta} = \frac{1}{m}\sum_{k=1}^{m}\hat{\theta}_k$$

$\hat{\theta}$ is the pooled point estimate, and $\hat{\theta}_k$ is the estimate from data set k.

For the pooled standard errors, both the within-imputation variance (that is based on a complete data set) and the between-imputation variance (that adds the uncertainty due to missing values) are combined. The within-imputation variance is the arithmetic average of the imputation's variances (standard errors squared):

$$\bar{V}_w = \frac{1}{m}\sum_{k=1}^{m} SE_k^2$$

The between-imputation variance is defined as the variance of point estimates between imputations that is due to uncertainty caused by the missing values:

$$V_b = \frac{1}{m-1}\sum_{k=1}^{m}(\hat{\theta}_k - \bar{\theta})^2$$

The total pooled standard error combines the within-imputation variance $\bar{V}_w$ and the between-imputation variance V_b:

$$SE = \sqrt{\bar{V}_w + V_b + \frac{V_b}{m}}$$

The third term in this expression acts as a correction factor for using a finite number of imputations in the Monte Carlo simulation (Enders, 2022, p. 284). As the number of imputations m increases, this correction term approaches zero. As the between-imputation variance is a measure of increase in uncertainty due to missing values, the influence of missing data on the variance of estimates can be expressed proportionally, e.g. as the relative increase of variance (RIV) due to missing values:

$$RIV = \frac{V_b + \frac{V_b}{m}}{\bar{V}_w}$$

With an appropriate point estimate and standard error, the remaining step (see Figure 2.2) is the choice of the sampling distribution. Rubin proposed a calculation method for degrees of freedom of the t-distribution, that is based on the RIV and the number of imputations m:

$$df_R = (m-1)\left(1 + \frac{1}{RIV^2}\right)$$

This formula assumes that the degrees of freedom of the complete data are effectively infinite (Barnard & Rubin, 1999). In smaller samples, the classic Rubin formula therefore tends to produce degrees of freedom that are too large, sometimes even larger than the degrees of freedom of the complete data. Barnard and Rubin [@-barnard] thus proposed a small-sample-adjusted degrees-of-freedom calculation for pooled parameters. Their formula downweights the complete-data degrees of freedom to account for the missing values. As the complete-data degrees of freedom cannot easily be obtained for multilevel analyses, a suitable degrees-of-freedom approximation must be chosen for the analysis. Possible approaches include the methods described in Section 2.3.1.3, such as the Kenward–Roger correction or the simple $m - l - 1$ rule.

2.4.2.2 Full Information Maximum Likelihood (FIML)

Full information maximum likelihood (FIML) is a model-based approach, as parameter estimates are obtained directly by maximizing the likelihood of the observed data (Enders, 2022, p. 98). Under the assumption that data are missing at random (MAR), FIML yields unbiased and efficient parameter estimates (D. McNeish, 2017a).

However, for the random intercept model with group-mean effects considered in this thesis, FIML is not always well suited when missingness occurs in predictor variables. In such cases, maximum likelihood estimation requires additional distributional assumptions about the predictors in order to incorporate them into the likelihood function (Grund et al., 2018). In multilevel models with manifest group means, these assumptions may lead to biased estimates of between-group effects (Grund et al., 2019).

For this reason, the focus of the present thesis lies on an alternative model-based approach for handling missing data: fully Bayesian estimation.

2.4.2.3 Fully Bayesian Estimation

With some additional specifications, the Bayesian estimation framework introduced in Section 2.3.2 can be readily applied to missing data problems.

With a complete dataset, the model of interest

$$p(y \mid X, \theta)$$

can be estimated directly. In this case, only the model parameters θ are unknown. Estimation relies on the observed relationship between X and Y. The result of the Bayesian analysis is the posterior distribution of the model parameters given the observed data:

$$p(\theta \mid y, X)$$

If missing values occur in predictor variables, the model of interest can be written as

$$p(y \mid X_{obs}, X_{mis}, \theta)$$

As only X_{obs} are available, the missing data X_{mis} are treated as additional unknown quantities, analogous to the model parameters θ (Gelman et al., 2025, p. 449). Consequently, inference is based on the joint posterior distribution of model parameters and missing values conditional on the observed data:

$$p(\theta, X_{mis} \mid X_{obs}, y)$$

Because the missing values are now quantities to be estimated, a model from which they can be drawn must be specified. If missings only occur in the outcome, this is equivalent to the model of interest. In contrast, when missing values occur in predictor variables, an additional model for the missing covariates must be specified. This implies specifying a joint model for all variables involved in the analysis, including both the outcome and the predictors:

$$p(y, X)$$

This joint distribution forms the basis for the algorithm used to generate imputations for the missing values. For example, a missing value in X can be imputed using the information contained in the other observed variables. A convenient way to specify this joint model is by making use of the probability chain rule to express it as a product of univariate distributions, which is called *sequential specification* or *factored regression specification* (Enders, 2022, p. 193). Advantages of such an equivalent specification are flexibility to account for a mixture of different metrics between variables (Erler et al., 2016) and the inclusion of auxiliary variables for imputation that are not required to be in the analysis model (Enders, 2022, p. 214). E.g., in the following equation, an observed variable W is included in the model to improve imputations of missing values, but the substantive model of interest is still estimated without W:

$$p(y, x, w) = p(w \mid y, x)\, p(y \mid x)\, p(x)$$

The procedure is usually carried out via a *Gibbs sampler*, a form of MCMC algorithm (Enders et al., 2016). This sampler alternates between drawing plausible values for the missing values and updating the model parameters conditional on these draws. These steps are repeated many times until the simulated distributions stabilize (Enders, 2022, p. 190). The so obtained missing data imputations are just a means to facilitate parameter estimation. Because in each iteration, all missing values are imputed newly, the additional uncertainty due to missing values is taken into account (Erler et al., 2016). Whereas in multiple imputation the Bayesian parameter inferences from the posterior distribution are discarded (Gelman et al., 2025, p. 451) and only the missing data imputations are saved, the parameter inferences are the end result in a fully Bayesian analysis.

Chapter 3

Methods

A simulation study was conducted in order to compare the performance of methods for missing data handling in small-sample multilevel random intercept models.
The idea behind a simulation study is to carry out a computer experiment (Morris et al., 2019), where data with known characteristics is simulated and subsequently analyzed with the methods of interest. The analysis results can be compared to the truth, allowing for assessment of the performance of the methods (Siepe et al., 2024).

Planning of the simulation study followed the ADEMP structure proposed by Siepe et al. (2024) and will be reported based on their guidelines.

3.1 Data generation mechanism

3.1.1 Data-generating model

The data-generating model is a two-level random intercept model:

$$Y_{ij} = \gamma_{10}(X_{ij} - \bar{X}_{\cdot j}) + \gamma_{01}\bar{X}_{\cdot j} + \gamma_{02}W_j + u_{0j} + e_{ij} \tag{3.1}$$

Y_{ij}, X_{ij} and W_j are parameterized to have a mean of zero and a variance of 1 on the population level.
First, between- and within-components of X_{ij} were simulated, independent of each other and with variances chosen to achieve a prespecified intraclass coefficient (see Section 3.1.3):

$$X_{ij} = u_{Xj} + e_{Xij},$$
$$u_{X_j} \sim N(0, {\tau_x}^2), \qquad e_{X_{ij}} \sim N(0, {\sigma_x}^2)$$

with ${\tau_x}^2 = ICC_x$ and ${\sigma_x}^2 = 1 - ICC_x$, respectively.
X_{ij} was then group-mean-centered.
The group-level covariate W_j was constructed with a linear regression equation to have a prespecified correlation with $\bar{X}_{\cdot j}$ ($\rho_{\bar{X}W}$):

$$W_j = a\bar{X}_j + e_{Wj} \qquad e_{Wj} \sim N(0, {\sigma_w}^2)$$

The residual variance ${\sigma_w}^2$ and the regression coefficient a are fully determined by the restriction that $Var(W_j) = 1$:

$$ {\sigma_w}^2 = 1 - \rho^2_{\bar{X}W},$$
$$a = \frac{\rho_{\bar{X}W}}{\sqrt{\tau_x^2 + \frac{{\sigma_x}^2}{n}}}$$

TABLE 3.1: Simulation conditions for the Data-Generating Model and the Generation of Missing Values

Data conditions	
N_1	10
N_2	15 / 30 / 60
ICC of X and Y	.1 / .3
Correlation $\bar{X}W$	.3
Model parameters	
Effect of $(X_{ij} - \bar{X}_{.j})$	.2
Effect of $\bar{X}_{.j}$	0 / .4
Effect of W_j	0
Missing values	
Pattern	$X \sim W$
Mechanism	MCAR / MAR
Proportion	.3

Finally, the random effects on the group and the individual level were generated as

$$u_{0j} \sim N(0, \psi^2), \qquad e_{ij} \sim N(0, \sigma^2)$$

and independent of each other.
Individual and group residual variances of Y_{ij} are determined by the constraints of a prespecified residual ICC of Y_{ij}, $Var(Y_{ij}) = 1$ and $Var(X_{ij}) = 1$[1]:

$$\psi^2 = ICC_{Y|X,W} \cdot (1 - {\gamma_{10}}^2(\sigma_X^2 - \frac{\sigma_X^2}{n_j}) + {\gamma_{01}}^2(\tau_x^2 + \frac{\sigma_X^2}{n_j})$$

$$\sigma^2 = (1 - ICC_{Y|X,W}) \cdot (1 - {\gamma_{10}}^2(\sigma_X^2 - \frac{\sigma_X^2}{n_j}) + {\gamma_{01}}^2(\tau_x^2 + \frac{\sigma_X^2}{n_j})$$

The full derivation of these equations is provided in **?@sec-appA**.

3.1.2 Missing data generation

After simulation of the complete dataset, values of the X-variable have been deleted to simulate missingness. A 'missingness tendency' r was simulated for each X-value according to

$$r_{ij} = \alpha + \beta W_j + e_{ij}, \qquad e_{ij} \sim N(0, 1 - \beta^2)$$

r_{ij} is normally distributed. Observations with $r_{ij} > 0$ were set to missing.
The intercept α controls the proportion of missingness in X, whereas the slope β determines the strength of the missing-at-random mechanism. When $\beta = 0$, missingness is completely at random (MCAR). With increasing β, the missingness mechanism becomes more dependent on the W-values; with $\beta = 1$, missingness would be completely deterministic.
For the present simulation, $\beta = 0.3$ was used for MAR conditions, corresponding to approximately 9% explained variance.

3.1.3 Factors of the data-generating mechanism

A summary of simulation conditions is provided in Table 3.1.

[1]as well as setting γ_{02} to be zero across all conditions, see Section 3.1.3

The number of Level-2 units was manipulated across conditions ($N_2 = 15, 30, 60$), while the number of Level-1 observations was kept constant ($N_1 = 10$). The between-cluster effect of $\bar{X}_{.j}$ was varied ($\gamma_{01} = 0, .40$), whereas the within-cluster effect of $(X_{ij} - \bar{X}_{.j})$ was held constant at .20 and the effect of W_j was set to zero. This results in conditions with either a negative and a positive context effect (i.e., $\gamma_{10} - \gamma_{01}$) of X in the population model. The ICC of X and the residual ICC of Y were varied ($\rho_X = \rho_{Y|X} = .10, .30$). Missing data were generated on X with a varying missingness mechanism (MCAR or MAR with a strength of $\beta = .30$), with a constant missingness rate of 30%. The factors were combined fully factorial, creating $3 \times 2 \times 2 \times 2 = 24$ conditions.
All in all, these settings reflect a number of the typical applications in cross-sectional and longitudinal applied research (e.g., smaller and larger intraclass correlation coefficients (ICC) and level-two sample sizes). They also create sufficient methodological variation to evaluate the properties of the methods of interest (e.g., null versus substantial effects).

3.2 Estimands of the focal analysis model

The focal analysis model is

$$Y_{ij} = \gamma_{10}(X_{ij} - \bar{X}._j) + \gamma_{01}\bar{X}._j + u_{0j} + e_{ij} \tag{3.2}$$

The primary target is the Level-2-effect, γ_{01}. Additionally, the Level-1-effect γ_{10} was evaluated.

3.3 Methods of Missing Data Handling

Four methods will be compared, alongside the benchmark analysis of Complete Data (CD).

3.3.1 Listwise Deletion (LD)

Listwise Deletion was used as a standard reference method for missing data handling. Cases with missing X-values were excluded prior to analysis. The group means $\bar{X}_{.j}$ were then calculated only from the remaining observations. Then, group-mean centering was applied.

3.3.2 Multiple Imputation (MI) with standard (MI-R) and adjusted (MI-a) degrees of freedom

For both multiple imputation methods, a joint modeling approach was employed. Although only values in X were missing, the multivariate specification preserves the multilevel structure during imputation. The multivariate empty model is generally suitable for random-intercep models (Grund et al., 2016):

$$\begin{pmatrix} Y_{ij} \\ X_{ij} \\ W_j \end{pmatrix} = \begin{pmatrix} \beta_{0Y} \\ \beta_{0X} \\ \beta_{0W} \end{pmatrix} + \begin{pmatrix} u_{Yj} \\ u_{Xj} \\ u_{Wj} \end{pmatrix} + \begin{pmatrix} e_{Yij} \\ e_{Xij} \\ 0 \end{pmatrix}$$

The random effects $(u_{Yj}, u_{Xj}, u_{Wj})^\top$ and the residuals $(e_{Yij}, e_{Xij})^\top$ follow multivariate normal distributions with mean zero and covariance matrices $\Psi = \begin{bmatrix} \psi_y^2 & \psi_{yx} & \psi_{yw} \\ \psi_{yx} & \psi_x^2 & \psi_{xw} \\ \psi_{yw} & \psi_{xw} & \psi_w^2 \end{bmatrix}$ and $\Sigma = \begin{bmatrix} \sigma_y^2 & \sigma_{yx} \\ \sigma_{yx} & \sigma_x^2 \end{bmatrix}$

W_j serves as an auxiliary variable to explain missingness.
Imputation was executed with the R package `jomo` (Quartagno & Carpenter, 2023) with 2000 burn-in-iterations and 3000 post-burn-in iterations, saving every 300th imputation (10 imputations).
For MI-R, degrees of freedom for calculating the test statistic were calculated with the software standard (Rubin´s rules, assuming infinite degrees of freedom in complete data).
For MI-a, degrees of freedom were calculated with $df_{\gamma_{10}} = M - r - 1$ (M: total number of Level-1 units; r: total number of predictors, in the present case 2) and $df_{\gamma_{01}} = N_2 - q - 1$ (N_2: number of Level-2 units; q: number of Level-2 predictors, here: 1)

3.3.3 Fully Bayesian Analysis

A factored regression specification was used within the Bayesian framework to allow the inclusion of the auxiliary variable W while leaving the focal analysis model unchanged:

$$g\left(y,\ x, w; \gamma\right) = g_w\left(w|y, x; \gamma_w\right)\ \times\ g_y\left(y|x; \gamma_y\right)\ \times\ g_x(x; \gamma_x) \tag{3.3}$$

Here, g_w is the conditional distribution of the auxiliary variable, while g_y corresponds to the substantive analysis model.
The burn-in period was set to 1000 iterations, followed by 3000 post-burn-in iterations. The model was estimated using the package `mdmb` (Robitzsch & Luedtke, 2024) with its default noninformative priors.

3.3.4 Extracted quantities

For complete data, listwise deletion and both multiple imputation methods, point estimates of γ_{10} and γ_{01} as well as their standard errors and 95% - confidence intervals were extracted.
For the fully Bayesian method, the posterior means of γ_{10} and γ_{01} were used as point estimates along with posterior standard deviations and 95% credible intervals. For comparability with the frequentist approaches and in line with common simulation practice, these posterior summaries were treated analogously to frequentist estimates and intervals (Morris et al., 2019).

3.4 Performance measures

Three performance measures were used for each parameter (γ_{10} and γ_{01}):

1. Bias of point estimates, defined as the difference between the parameter´s mean estimated and true value.
2. Coverage of confidence/credible intervals, defined as the proportion of simulation repetitions that captured the true parameter value inside the CI.
3. Standard deviation of the estimations across simulation repetitions.

Monte Carlo uncertainty of the estimated performance measures was calculated using Monte Carlo Standard Errors (MCSE), using the formulas provided in Siepe et al. (2024).
For each condition, 1000 simulation replications were used, based on comparable simulation studies (Enders et al., 2016; Grund et al., 2018) and ensuring a sufficiently small MCSE of $< 1\%$ for coverage [2].

The simulation was conducted in RStudio with the aim of full reproducibility, following guidelines of the repro book (ZITIEREN), e.g. by setting a seed for simulation and using reproducible package environments. The code and all relevant information for reproduction can be found in the digital version of the master thesis. (*LINK*)

Noch einfügen: wie Performance measures evaluiert werden sollen; außerdem width of confidence intervals als Performance measure?

[2] MCSE calculation for bias and standard deviation are based on the sample variance of the estimates and cannot be pre-calculated.

Chapter 4

Results

The complete results for the parameters of interest are provided in tables in **?@sec-appB**.

4.1 Bias of point estimation

In line with expectations, no substantial bias was observed for the individual-level effect γ_{10}.
The findings for γ_{01} are depicted in Figure 4.1. For γ_{01}, listwise deletion produced substantial bias, especially in low-ICC conditions, with positive bias for the null effect and negative bias for the positive between-group effect. With increasing sample size and under MAR condition, this bias was more pronounced.
Multiple imputation showed even larger bias than listwise deletion in conditions with low ICC and positive effect[1]. In contrast, the Bayesian approach only led to noticeable bias in the most challenging conditions (smallest sample size, low ICC and positive γ_{01}-effect).

4.2 Coverage of confidence/credible intervals

The coverage of the confidence and credible intervals for γ_{01} is shown for $N_2 = 15$ in Figure 4.2, as coverage for larger sample sizes was largely within the acceptable range of 0.925–0.975.

Listwise deletion yielded coverage rates close to the nominal 95% level across all conditions. For multiple imputation, the small-sample adjusted degrees of freedom resulted in increased overcoverage compared to the standard degrees of freedom, particularly when $N_2 = 15$, with a similar but weaker tendency at larger sample sizes. The Bayesian approach showed even stronger overcoverage in the smallest sample, but not for $N_2 = 30$ and $N_2 = 60$.

Overall, MI with standard degrees of freedom (MI-R) and listwise deletion provided coverage closest to the nominal level, whereas noticeable deviations occurred mainly in the smallest Level-2 sample for the other methods.

[1] However, in a sensitivity analysis, usage of alternative weakly informative priors in these conditions consistently reduced bias by roughly 30–60% (see **?@sec-appC** for details).

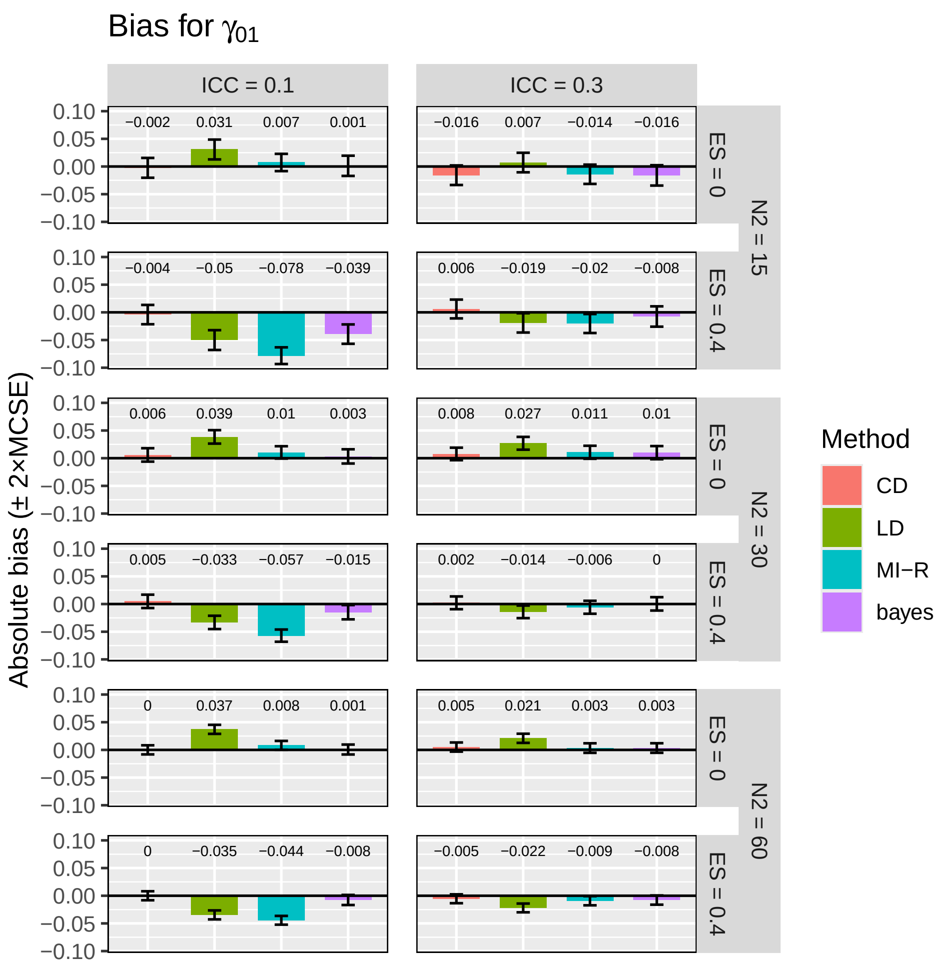


FIGURE 4.1: Hier noch Caption

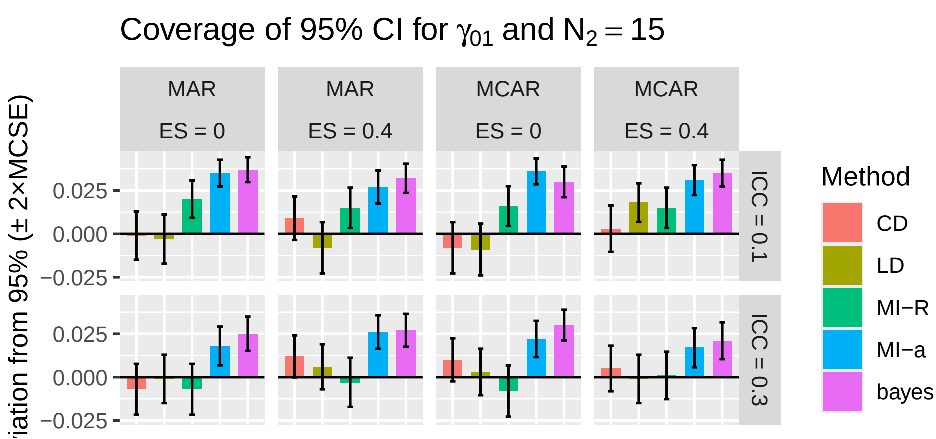


FIGURE 4.2: Coverage TEXT TEXT TEXT HIER HIN NOCH TEXT

4.3 Standard deviation of point estimates

No substantial differences between methods for the parameter γ_{10} were observed, except for the complete-data condition, which consistently yielded the smallest standard deviations.
Interestingly, for γ_{10}, the variability of point estimates across simulation repetitions was smallest for multiple imputation in low-ICC conditions and similar to complete data analysis for high-ICC, whereas the Bayesian approach produced the largest standard deviations across all conditions (see Figure 4.3). A systematic tendency was visible: Multiple imputation standard deviations decreased with a low ICC, whereas in the Bayesian approach, they decreased with a high ICC for $N_2 = 30$ and $N_2 = 60$. Listwise deletion shows very similar results to complete data analysis regarding standard deviation.
Differences between methods are overall quite small, with MI SD being smaller than Bayes SD by a maximum of ~14.5%.

Hier eher Tabelle statt Grafik?

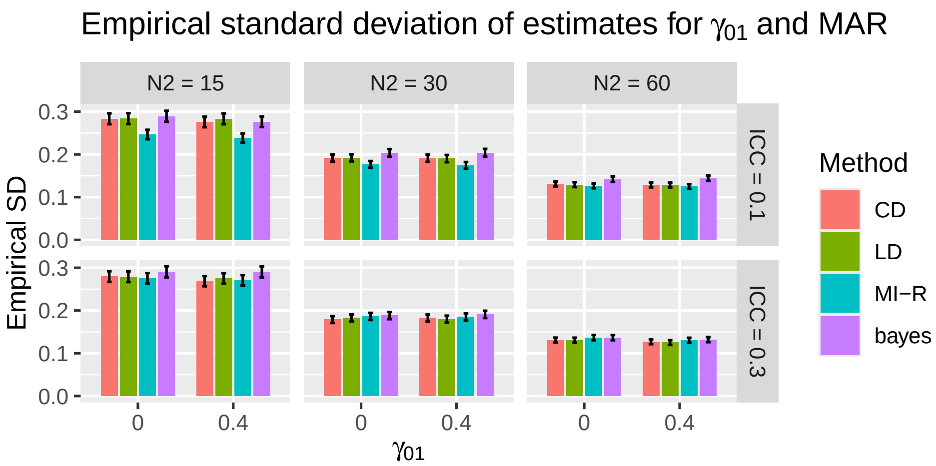


FIGURE 4.3: blablabalbslsd

4.4 Width of confidence / credible intervals

No noteworthy differences were observed for γ_{10}.
For γ_{01}, interval widths were relatively stable within each sample size, but there were differences between methods. Bayesian credible intervals were consistently the widest, followed by the MI-a approach. Confidence intervals from the MI with Rubin´s degrees of freedom were even narrower than CD/LD for $N_2 = 15$, although these differences were small in magnitude.
These results are shown in an exemplary manner in Figure 4.4.

4.5 Summary of results

Across all performance criteria, only small differences between methods were observed for the individual-level effect γ_{10}.
Regarding γ_{01}, listwise deletion showed increased bias but acceptable coverage. Multiple imputation led to higher bias than the Bayesian approach in several challenging

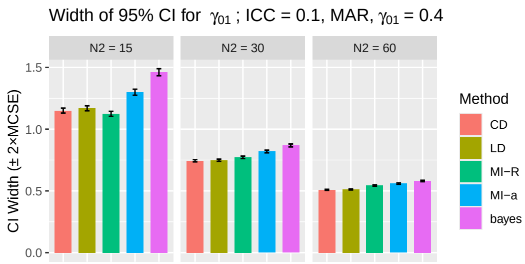


FIGURE 4.4: blablabalbslsd

conditions. MI-a produced higher overcoverage and wider confidence intervals compared to the MI-R variant; the Bayesian method generally resulted in the highest overcoverage and the widest credible intervals.

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